HOST_POWER_MANAGEMENT

Host Power Management — Hard Ban (CONST-033)

Why this exists

On 2026-04-26 18:23:43 the host running mission-critical parallel CLI agents and container workloads was auto-suspended mid-session. This killed the running HelixAgent binary, all 41 dependent services, every SSH session, and every active CLI agent on the box. journalctl showed:

```
systemd-logind[1183]: The system will suspend now!
```

The user-level GNOME power settings were already correct (`sleep-inactive-ac-type=nothing`). The trigger was the **GDM greeter session at the local console**, which has its own power policy and does not count SSH activity. Earlier, on multiple occasions, the user@1000.service had been SIGKILLed by systemd because heavy memory pressure prevented gnome-shell from responding to GDM/Wayland watchdog within `TimeoutStopSec` — perceived by the user as "the system fully logged me out." Together these two failure modes have caused repeated loss of in-flight agent work.

The rule

No project shipped from this workspace may invoke a host-level power-state transition. Forbidden invocations include — but are not limited to:

Layer	Forbidden invocations
systemd CLI	`systemctl suspend`, `systemctl hibernate`, `systemctl hybrid-sleep`, `systemctl suspend-then-hibernate`, `systemctl poweroff`, `systemctl halt` `systemctl reboot`, `systemctl kexec`
logind CLI	`loginctl suspend`, `loginctl hibernate`, `loginctl hybrid-sleep`, `loginctl` `suspend-then-hibernate`, `loginctl poweroff`, `loginctl halt`, `loginctl reboot`
Legacy CLI	`pm-suspend`, `pm-hibernate`, `pm-suspend-hybrid`, `shutdown -h/-r/-P/-H/now` bare `reboot/poweroff/halt`
DBus	`org.freedesktop.login1.Manager.{Suspend,Hibernate,HybridSleep,SuspendThenHibernate,PowerOff,Reboo` `org.freedesktop.UPower.{Suspend,Hibernate,HybridSleep}` (via `dbus-sen` `busctl`, or any language binding)
gsettings	`gsettings set ... sleep-inactive-{ac,battery}-type` to anything other than `'nothing'` or `'blank'`

Defence in depth

Three layers, in order of strength:

1. **Host-level masking (manual prereq, sudo required, run once):** `scripts/host_power_management/install-host-suspend-guard.sh` masks `sleep.target`, `suspend.target`, `hibernate.target`, `hybrid-sleep.target`, writes `/etc/systemd/sleep.conf.d/00-no-suspend.conf` with `AllowSuspend=no`, and writes `/etc/systemd/logind.conf.d/00-no-idle-suspend.conf` with `IdleAction=ignore` and `HandleLidSwitch=ignore`. After this, no user / session / DE / greeter / cron job can suspend the host.

2. **User-session bootstrap (no sudo):** `scripts/host_power_management/user_session_no_suspend_bootstrap.sh` runs `gsettings`, `xset -dpms`, and (opt-in via `HOST_POWER_MANAGEMENT_SESSION_INHIBIT=1`) `systemd-inhibit` to protect the current GUI/CLI session of the invoking user. Idempotent. Safe to source from `start.sh` / `setup.sh` / `bootstrap.sh`.

3. **Source-tree static gate:** `scripts/host_power_management/check-no-suspend-calls.sh` walks the tree and exits non-zero on any forbidden invocation. `scripts/anti_bluff/no_suspend_calls_challenge.sh` wraps it as a challenge that runs in CI / `run_all_challenges.sh`. `scripts/anti_bluff/host_no_auto_suspend_challenge.sh` asserts the running host's state matches the layer-1 masking.

Verification

```bash
# Layer 1: host state
bash scripts/anti_bluff/host_no_auto_suspend_challenge.sh
# Expected: 4 PASS

# Layer 3: source tree
bash scripts/anti_bluff/no_suspend_calls_challenge.sh
# Expected: PASS, no forbidden calls
```

If you genuinely need a power-state transition

You don't, in this workspace. If a future legitimate use case arises (e.g. a container's *internal* init script that suspends a *guest VM*, not the host), add the specific file path to `EXCLUDE_PATHS` at the top of `check-no-suspend-calls.sh` with a comment explaining the non-host context.

Sources verified 2026-05-29: https://www.freedesktop.org/software/systemd/man/latest/logind.conf.html , https://www.freedesktop.org/software/systemd/man/latest/org.freedesktop.login1.html

Verified against latest official systemd/logind documentation on 2026-05-29. The logind.conf drop-in mechanism (`/etc/systemd/logind.conf.d/*.conf`), the inhibitor/power-state surface (`loginctl suspend|hibernate|hybrid-sleep|suspend-then-hibernate|poweroff|halt|reboot`), and the D-Bus `org.freedesktop.login1.Manager` Suspend/Hibernate/HybridSleep/PowerOff/Reboot methods all match the current freedesktop.org systemd manuals. This doc enumerates the FORBIDDEN host-power-management surface per CONST-033 (hard ban) and is descriptive of the ban, not an instruction to invoke any of these — it correctly does NOT instruct any power-state transition.

Negative findings: none — the document is a constitutional ban reference, not an operational guide, and the cited systemd interfaces remain accurate.